Yav Rohatgi

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Education

University of Massachusetts Amherst — Amherst, USA

Bachelor of Science in Computer Engineering (Honors), Minor in Mathematics; GPA: 3.82/4.0 — *Sep 2021 – May 2025*

- Honors: Chancellor's Award (Merit Scholarship, $48000) and Dean's List (every semester)
- Thesis: AI Assisted Approach to Math Tutoring advised by Professors Lixin Gao & Andrew Lan
- Coursework: Artificial Intelligence, Machine Learning, Data Structures and Algorithms, Secure Distributed Systems, Operating Systems, Computer Networks, Linear Algebra, Probability, Statistics & Calculus

Work Experience

Software Engineer — Jun 2025 – Present

Sonet.io — *Gurgaon, India*

- Developed an agentic AI solution using browser automation to read and drive web based application screens, handling logins, data entry, and workflow execution, achieving 80% accuracy on PMS workflows
- Automated VM image preparation using CI/CD pipelines, including WinRM validation, sysprep, VM generalization, and Shared Image Gallery (SIG) publishing, streamlining and speeding up image release cycles
- Built custom browser automation actions using Chrome DevTools Protocol for precise coordinate based clicking and typing on canvas based UIs, enabling reliable automation on web applications, where DOMs fail

Software Engineer Intern — May 2024 – Aug 2024

Sonet.io — *Gurgaon, India*

- Cut manual setup time by 7% by engineering PowerShell scripts to automate installs on 50+ remote machines
- Increased log access speed by 8% by building Kubernetes monitoring scripts for 10+ applications across 3+ clusters
- Improved system and storage efficiency by 5% through logout scripts that deleted guest-created data

Software Engineer Intern — May 2023 – Aug 2023

Samsung SDS — *Gurgaon, India*

- Boosted sentiment classification accuracy by 25% by developing a Python program for 50K+ product reviews
- Customized the VADER library to identify key product features tied to market shifts across 100+ smartphones
- Delivered insights to product teams that influenced market strategy and product improvement decisions

Undergraduate Teaching Assistant — Sep 2023 – Dec 2024

University of Massachusetts, Amherst — *Amherst, USA*

- Helped 150+ students by conducting weekly office hours and debugging their Python & MATLAB assignments
- Assisted students by breaking down assignment problems into clear steps and explaining the question
- Handled course logistics such as grading, student coordination, and communicating common issues to instructor

Projects

AI Assisted Approach to Math Tutoring | *LLM Fine-Tuning, Prompt Engineering, RAG* — Sep 2024 – May 2025

- Achieved 90% accuracy on 5,000+ math problems by integrating N-shot prompting, Chain-of-Thought reasoning, and symbolic reasoning using Wolfram Alpha, which improved multi step reasoning
- Deployed a RAG system that helped improve BLEU by 11.6% for lecture-style question generation
- Discovered that 18% of generated problems were unsolvable, revealing internal failures in LLM reasoning

Sign Language Detection Gloves | *Machine Learning, Python, C++, TensorFlow, IMUs* — Sep 2024 – May 2025

- Captured high-frequency motion signals from 6-axis IMU sensors on each finger to enable detailed gesture inputs
- Processed motion data into discrete-time series timestamped via MCU clock to recognize sequential hand signs
- Reached 92% real-time classification accuracy by training a lightweight model with TensorFlow Lite

Compact LLM Architecture | *Machine Learning, JAX* — Sep 2024 – Dec 2024

- Built sequence prediction models (constant, linear, MLP, double-layer) to evaluate accuracy and tradeoffs
- Reduced loss by 66% by fine-tuning double MLP networks with SGD and improved text clarity

Technical Skills

Languages: Python, C, SQL, PowerShell, Bash
Developer Tools: Docker, Kubernetes, Git & GitHub, Azure, AWS
Libraries: TensorFlow, PyTorch, JAX, Scikit-Learn, Hugging Face, LangChain, Pandas, NumPy, Matplotlib